

Multimodal Emotion Recognition Using Transformer-Based Fusion on the MELD Dataset

Abstract

Emotion recognition is a critical component of human-computer interaction systems. This study presents a multimodal emotion recognition framework that integrates textual and visual information using Transformer-based architectures and convolutional neural networks. Initially, independent image and text datasets were explored; however, lack of alignment between modalities resulted in poor performance.

To address this limitation, the Multimodal EmotionLines Dataset (MELD) was adopted, enabling synchronized multimodal learning. The proposed model employs a Transformer encoder for textual data and a CNN-based encoder for visual features, followed by a fusion mechanism for classification. Experimental results demonstrate improved performance of multimodal learning over unimodal baselines, achieving an accuracy of approximately 42% and ROC-AUC of 0.73. The study highlights challenges in dataset alignment, class imbalance, and multimodal fusion, and proposes directions for further improvement.

I. Introduction

Emotion recognition has become an essential task in artificial intelligence, particularly in applications such as conversational agents, mental health monitoring, and human-computer interaction. Traditional approaches relied on unimodal data such as text or facial expressions, which often fail to capture the complexity of human emotions.

Recent advances in deep learning enable the integration of multiple modalities, improving model robustness and interpretability. This project focuses on building a multimodal emotion recognition system combining textual and visual inputs using modern deep learning techniques, particularly Transformers and convolutional neural networks.

II. Problem Statement

The primary objective of this project is to:

- Develop a multimodal emotion detection system
- Combine textual and facial expression data effectively
- Improve classification performance over unimodal models
- Address real-world challenges such as class imbalance and data misalignment

III. Related Work

Recent research highlights the effectiveness of Transformer-based architectures in natural language processing and multimodal tasks. Models such as BERT and its variants have shown strong performance in text classification tasks, while CNNs remain effective for image feature extraction.

Multimodal datasets like MELD provide synchronized data across modalities, enabling more meaningful fusion strategies. Prior work indicates that multimodal approaches outperform unimodal systems, particularly when modalities are properly aligned.

IV. Dataset and Preprocessing

A. Initial Dataset Attempt (Challenges)

Initially, separate datasets were used:

- Facial emotion dataset (images)
- Text emotion dataset (CSV)

Challenges faced:

- No alignment between text and images
- Random pairing led to meaningless learning
- Model performance stagnated (~30–38%)
- Multimodal fusion ineffective

Key insight:

Multimodal learning requires paired and synchronized data

B. Final Dataset: MELD

The MELD dataset provides:

- Text (dialogues)
- Video (facial expressions)
- Emotion labels

Preprocessing steps:

1. Extract video frames using FFmpeg
2. Map frames to utterances
3. Tokenize text using Transformer tokenizer

4. Generate attention masks
5. Align image-text pairs per sample

V. Methodology

A. Model Architecture

The system consists of three main components:

1. Text Encoder

- Transformer-based (DistilBERT/BERT)
- Input: tokens + attention mask
- Output: contextual embeddings (CLS token)

2. Image Encoder

- CNN-based feature extractor
- Input: facial frame
- Output: visual feature vector

3. Multimodal Fusion

- Concatenation of features
- Optional cross-modal attention
- Fully connected classification layers

B. Training Strategy

- Loss Function: Cross-Entropy (with class weights)
- Optimizer: AdamW
- Scheduler: Cosine annealing
- Regularization: Dropout

VI. Experimental Setup

- Hardware: NVIDIA H100 GPU
- Framework: PyTorch
- Text Model: DistilBERT
- Batch Size: 8–16
- Epochs: 10

VII. Results and Discussion

A. Multimodal Performance

Metric	Value
Accuracy	~42%
ROC-AUC	~0.73
Top 2 Accuracy	~72%

B. Observations

- Multimodal model outperformed unimodal approaches
- Strong improvement after switching to MELD dataset
- ROC-AUC indicates good ranking ability

C. New Unimodal Image Model

To further investigate the impact of stronger visual feature extract, a ResNet-18 model pretrained on ImageNet was implemented for image modality. Unlike the earlier CNN encoder train from scratch on a limited dataset (~10,00 images), the pretrained ResNet-18 leverages large scale feature representations learned from over 1 million images.

This approach resulted in a significant performance improvement, achieving approximately 80% test accuracy, compared to ~55% using the earlier CNN model. Additionally, validation loss decreased consistently during training, indicating improved generalization.

The classification report shows strong performance across most classes, with particularly high precision and recall for dominant classes such as Happy and Neutral. Importantly, this model successfully learned classes that were previously not captured by earlier models.

These results highlight the importance of transfer learning and pretrained architectures when working with limited datasets. This model has yet to be integrated into the multimodal framework and serves as a strong unimodal baseline for future multimodal improvements.

D. Class-wise Issues

Class	Issue
disgust	Not learned
fear	Not learned
sadness	Overpredicted
neutral	Underrepresented

Cause: **class imbalance**

VIII. Key Challenges

1. Dataset Misalignment

- Initial datasets were independent
- No meaningful pairing between modalities
- Led to poor multimodal learning

2. Pipeline Integration Issues

- Transition from GRU → Transformer caused incompatibilities
- Errors due to mismatched inputs (text_lengths vs attention_mask)
- Required full pipeline restructuring

3. Frame Extraction Challenges

- Incorrect directory structure
- Missing dev/test frames
- Required debugging of MELD processing

4. Class Imbalance

- Rare classes underrepresented
- Model biased toward dominant classes

IX. Progress Achieved

This project demonstrates significant progress:

Phase 1

- Basic GRU-based multimodal system
- Low performance and unstable results

Phase 2

- Debugging dataset issues
- Understanding multimodal alignment

Phase 3

- Migration to Transformer-based architecture
- Full pipeline redesign

Phase 4

- Successful MELD integration
- Stable training and evaluation

Final system:

- Fully functional
- Research-level pipeline
- Multimodal + Transformer-based

X. Future Work

- Address class imbalance using Focal Loss and sampling
- Use video sequence modeling instead of single frame
- Incorporate audio modality
- Improve fusion using attention/gating mechanisms
- Apply GAN-based data augmentation
- Integrate the pretrained ResNet-18 encoder

XI. Conclusion

This study presents a Transformer-based multimodal emotion recognition system using the MELD dataset. While multimodal learning is generally expected to outperform unimodal approaches, the current implementation does not yet surpass the strong pretrained unimodal image baseline. This is likely due to the relatively weak CNN based image encoder (~55% accuracy), which limits the quality of visual feature representations available for fusion. Despite challenges such as class imbalance and pipeline complexity, the model achieves competitive performance and establishes a strong foundation for future improvements.

References

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3. Vaswani et al., “Attention is All You Need,” 2017